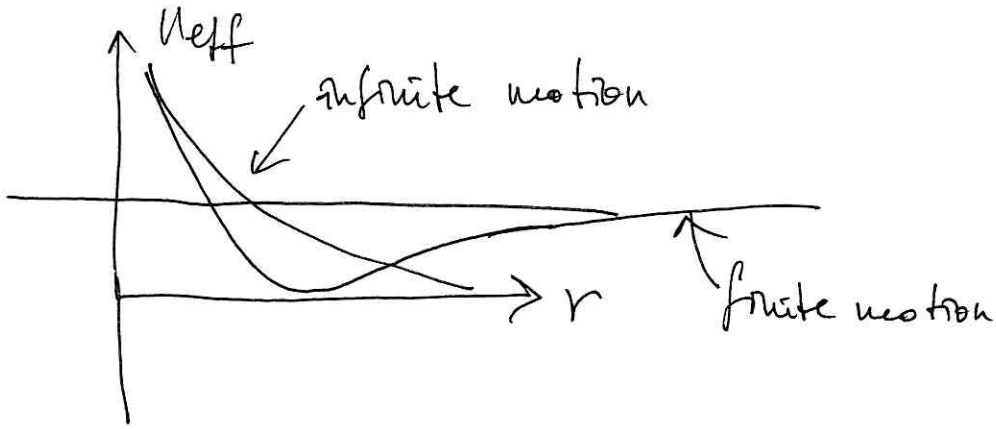
(P1) Motion in the central field is characterized by the effective potential

$$U_{eff} = U(r) + \frac{M^2}{2mr^2}$$

So that $E = \frac{1}{2} m \dot{r}^2 + U_{eff}(r)$.

Hence motion will be finite if $U_{eff}(r)$ has a <u>minimum</u> at some r.



(a) $U(r) = -Ve^{-\beta^2 r^2}$

$$\Rightarrow \frac{\partial}{\partial r} U_{eff} = 0 = \frac{\partial}{\partial r}\left(\frac{M^2}{2mr^2} - Ve^{-\beta^2 r^2} \right) \Rightarrow$$

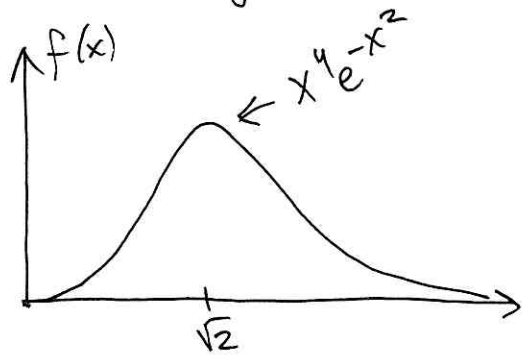
$$\Rightarrow -\frac{M^2}{mr^3} + 2\beta^2 V e^{-\beta^2 r^2} \cdot r = 0$$

$$r^4 e^{-\beta^2 r^2} = \frac{M^2}{2\beta^2 V m}$$

Introduce $x = \beta r$, then

$$x^4 e^{-x^2} = \frac{\beta^2 M^2}{2Vm} \quad (1)$$

Equation (1) has left-hand side which is function of $x = \beta r$, while the right-hand side is constant. The two can be equal only if $\frac{\beta^2 M^2}{2Vm}$ is smaller than the <u>maximum</u> value of the right-hand side: thus study the behavior of $f(x) = x^4 e^{-x^2}$



$$f'(x) = 4x^3 e^{-x^2} - 2x^5 e^{-x^2} = 0$$

$$(2 - x^2) 2x^3 e^{-x^2} = 0$$

$$x = \sqrt{2}$$

$$f(\sqrt{2}) = (\sqrt{2})^4 e^{-2} = 4e^{-2}$$

$$\Rightarrow \quad \frac{\beta^2 M^2}{2mV} \leq \frac{4}{e^2} \quad \Rightarrow \quad \boxed{M \leq \left(\frac{2}{e}\right)\sqrt{\frac{2mV}{\beta^2}}}$$

Condition on finite motion.

(b) $V(r) = -\frac{\alpha}{r} e^{-\beta r}$

$$\frac{\partial U_{eff}}{\partial r} = 0 = -\frac{M^2}{mr^3} + \frac{\alpha e^{-\beta r}}{r^2} + \frac{\alpha \beta e^{-\beta r}}{r} = 0$$

$$(\alpha r + \alpha \beta r^2)\, e^{-\beta r} = \frac{M^2}{m}$$

$$(1 + \beta r)\, r e^{-\beta r} = \frac{M^2}{m\alpha}$$

let $x = \beta r$: $\underbrace{(1+x)\, x e^{-x}}_{f(x)} = \frac{\beta M^2}{\alpha m}$

$$f'(x) = (1+2x)\, e^{-x} - (1+x)\, x e^{-x} = 0 \Rightarrow x^2 - x - 1 = 0$$

(take positive root) $x = \frac{\sqrt{5}+1}{2}$

$$f\left(\frac{\sqrt{5}+1}{2}\right) = \frac{\sqrt{5}+1}{2}\,\frac{\sqrt{5}+3}{2}\,e^{-\frac{\sqrt{5}+1}{2}} \cong 0.84$$

$\Rightarrow \quad M \leq \sqrt{0.84\,\frac{\alpha m}{\beta}}$ for the motion to be finite.

(P2) $L = \frac{1}{2} m \dot{\vec{r}}^2 - U(r) = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\varphi}^2 - U(r)$

(a) $\frac{d}{dt} \frac{\partial L}{\partial \dot{\varphi}} = 0 \Rightarrow m r^2 \dot{\varphi} = M = \text{const}$

also, $E = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\varphi}^2 + U(r) =$

$$= \frac{1}{2} m \dot{r}^2 + \underbrace{\frac{M^2}{2mr^2} + U(r)}_{U_{eff}(r)} = \text{const.}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} = \frac{\partial L}{\partial r}$$

$$m \ddot{r} = m r \dot{\varphi}^2 - \frac{\partial}{\partial r} U(r) = m r \left(\frac{M}{mr^2}\right)^2 - U'(r) =$$

$$= \frac{M^2}{mr^3} - U'(r)$$

$$\Rightarrow m \ddot{r} = -\frac{\partial}{\partial r}\left(\frac{M^2}{2mr^2} + U(r)\right) = -\frac{\partial}{\partial r} U_{eff}(r). \quad (1)$$

(b) $\frac{d\varphi}{dt} = \frac{M}{mr^2} \Rightarrow t = \frac{mr^2}{M} \varphi + c$

Hence $\dot{r} = \frac{dr}{dt} = \frac{dr}{d\varphi} \frac{d\varphi}{dt} = \frac{M}{mr^2} \frac{dr}{d\varphi}$

$$\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{M}{mr^2} \frac{d}{d\varphi}\left(\frac{M}{mr^2} \frac{dr}{d\varphi}\right) =$$

$$= \left(\frac{M}{mr^2}\right)^2 \frac{d^2 r}{d\varphi^2} - \frac{2M^2}{m^2 r^5}\left(\frac{dr}{d\varphi}\right)^2$$

Eq. (1) becomes: $\frac{M^2}{m r^4}\left(\frac{d^2 r}{d\varphi^2} - \frac{2}{r}\left(\frac{dr}{d\varphi}\right)^2\right) = \frac{M^2}{mr^3} - \frac{dU}{dr}$; (2)

(c) let $r(\varphi) = r_0 e^{-\alpha\varphi} \Rightarrow \frac{dr}{d\varphi} = -\alpha r_0 e^{-\alpha\varphi} = -\alpha r(\varphi)$

$$\frac{d^2r}{d\varphi^2} = \alpha^2 r_0 e^{-\alpha\varphi} = \alpha^2 r(\varphi).$$

Eq. 2 becomes: $\frac{M^2}{m r^4}\left[\alpha^2 r - \frac{2}{r}(-\alpha r)^2\right] = \frac{M^2}{m r^3} - \frac{dU}{dr}$

$$(\alpha^2 r - 2\alpha^2 r = -\alpha^2 r)$$

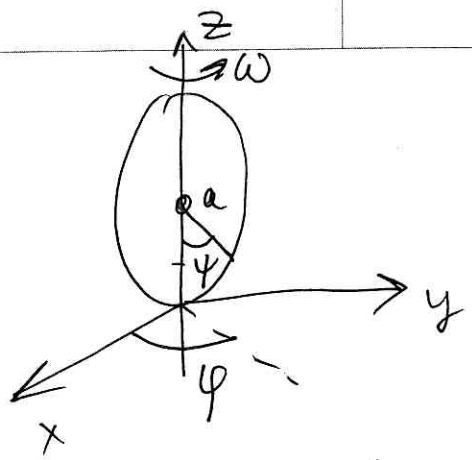
$$\Rightarrow \frac{dU}{dr} = \frac{M^2}{m r^3} + \frac{M^2}{m r^4}(\alpha^2 r) = \frac{M^2}{m}\left(1+\alpha^2\right)\frac{1}{r^3}$$

$$\Rightarrow U(r) = \frac{-M^2}{2m}\left(1+\alpha^2\right)\frac{1}{r^2} + \text{Const}$$

physical potentials satisfy $U(r\to\infty) = 0 \Rightarrow \text{Const} = 0.$

$$U(r) = -\frac{(1+\alpha^2)M^2}{2mr^2}$$ for the logarithmic spiral to realize.

P3



$$x = a\sin\psi\cos\varphi$$
$$y = a\sin\psi\sin\varphi$$
$$z = -a\cos\psi + a$$

with $\varphi = \omega t$.

(a)
$$T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2) = \frac{1}{2}ma^2\Big[(\cos\psi\cos\varphi\,\dot{\psi} - \sin\psi\sin\varphi\,\dot{\varphi})^2$$
$$+(\cos\psi\sin\varphi\,\dot{\psi} + \sin\psi\cos\varphi\,\dot{\varphi})^2 + (+\sin\psi\,\dot{\psi})^2\Big] =$$
$$= \frac{1}{2}ma^2\Big[\cos^2\psi(\cos^2\varphi+\sin^2\varphi)(\dot{\psi})^2 + \sin^2\psi(\sin^2\varphi+\cos^2\varphi)(\dot{\varphi})^2$$
$$+\sin^2\psi\,(\dot{\varphi})^2\Big] = \frac{1}{2}ma^2\left[\omega^2\sin^2\psi + \dot{\psi}^2\right]$$

where we used $\dot{\varphi} = \omega$.

$$U = mgz = mga(1-\cos\psi)$$
$$L = T - U = \frac{1}{2}ma^2(\dot{\psi}^2 + \omega^2\sin^2\psi) + mga\cos\psi - mga$$

can be droped.

(b)
$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\psi}} = \frac{\partial L}{\partial\psi} = -\frac{\partial}{\partial\psi}\underbrace{\left(-mga\cos\psi - \frac{1}{2}ma^2\omega^2\sin^2\psi\right)}_{V_{eff}(\psi)}$$

$$ma^2\ddot{\psi} = -\frac{\partial}{\partial\psi}V_{eff}(\psi)$$

Stationary points: $\frac{\partial V}{\partial\psi} = 0$

$$\Rightarrow mga\sin\psi - ma^2\omega^2\sin\psi\cos\psi = 0$$

$$\sin\psi\left(1-\frac{a\omega^2}{g}\cos\psi\right)=0 \Rightarrow \psi_1=0 \text{ or } \cos\psi_2=\frac{g}{a\omega^2}$$

Min or max?

$$\frac{\partial^2 V_{eff}}{\partial\psi^2} = mga\cos\psi - ma^2\omega^2(\cos^2\psi-\sin^2\psi) =$$

$$= mga\left(\cos\psi - \frac{a\omega^2}{g}\cos 2\psi\right)$$

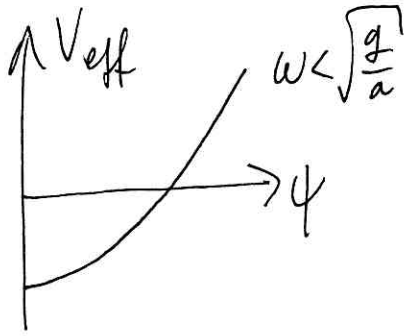
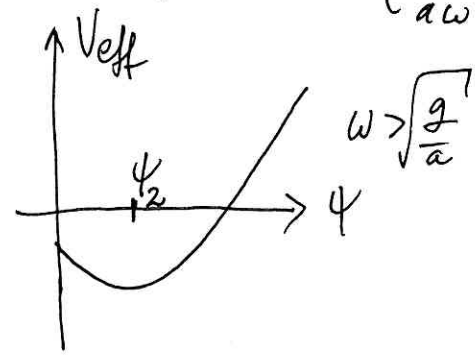
$$\frac{\partial^2 V_{eff}(\psi_1)}{\partial\psi^2} = mga\left(1-\frac{a\omega^2}{g}\right) > 0 \text{ for } 1 > \frac{a\omega^2}{g}$$

$$\Rightarrow \text{for } \omega < \sqrt{\frac{g}{a}}.$$

$\Rightarrow \psi=0$ is min for $\omega<\sqrt{\frac{g}{a}}$.

$$\frac{\partial^2 V_{eff}(\psi_2)}{\partial\psi^2} = mga\left(\frac{g}{a\omega^2} - \frac{a\omega^2}{g}\left(2\cdot\left(\frac{g}{a\omega^2}\right)^2-1\right)\right) =$$

$$= mga\left(\frac{a\omega^2}{g} - \frac{g}{a\omega^2}\right) = mga\left(\frac{g}{a\omega^2}\right)\left[\left(\frac{a\omega^2}{g}\right)^2-1\right]$$

$$>0 \text{ for } \frac{a\omega^2}{g}>1 \Rightarrow \omega>\sqrt{\frac{g}{a}}$$

Thus for high frequency of rotation, $\omega>\sqrt{\frac{g}{a}}$, the minimum at $\psi_2=\cos^{-1}\left(\frac{g}{a\omega^2}\right)$ appears.